CS6570: Secure System Engineering

Assignment 3: Format String Vulnerability (100 Marks)

You are provided with a Dockerfile containing a binary that needs to be exploited. Your task is to perform the following:

a) Similar to the previous assignment, there exists a secret message stored in the variable `secret` and a hidden function called `flag`. You are required to read the value stored in the variable and invoke the `flag` function by exploiting the vulnerabilities present in the program.

The program checks for a `.txt` file in the current directory and prints the file name along with its contents.

Task 1: Generate a payload named *payload_1* that can **read the content stored in the variable** `secret`. (20 marks)

b) Using the format string and buffer overflow vulnerabilities, generate a payload named *payload_2* that can **invoke the hidden function** `flag`. (30 marks)

A **clean exit** after the execution of each payload carries **10 marks**.

The report explaining the payload generation process carries **40 marks**.

Submission Guidelines

Submit a zipped file named:

`<your_roll_no.>.zip`

The zip file must contain:

1. *payload_1*
2. *payload_2*
3. Scripts used to generate the corresponding payloads
4. A detailed report explaining the payload generation process